LIBBY — BIOMARKER SURVEY

ulms-highgrade-pulm-mets-tp53-smarcb1-hla-a2-w9t4

Selected general biomarker report — biomarker screening coverage and gaps

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Selected general biomarker report

Which biomarkers on the selected panel this patient has been tested for, and which have never been measured. The panel covers targets established in this tumor type, targets relevant across a subset of tumor types, and biomarkers that predict benefit regardless of where the tumor started. An unmeasured biomarker below is a gap in the record, not a prediction that the patient carries it.

Surveyed: 15 biomarkers. **No usable result on file:** 12. **Measured but not decision-grade:** 2. **Already established:** 1.

Fifteen biomarkers were surveyed, and the source case's headline, a molecular landscape entirely unmeasured, no longer holds: a Guardant plasma ctDNA assay and HLA class I typing have since been done, and the gap narrowed rather than closed. Twelve were never measured, two measured short of decision grade, one settled. Four gaps are essential: MSI/dMMR, TMB, NTRK fusion, and HR/BRCA status.

The Guardant tumor fraction was never supplied, so its non-detections exclude nothing: a plasma negative is not a tissue negative. NTRK and HR/BRCA rest on plasma non-detections that only an RNA fusion panel and a tissue HRD score with germline testing can harden. MSI/dMMR and TMB stayed not measured, deliberately: the plasma assay returned no call, so there is nothing to harden.

HLA typing is settled; typing is not eligibility. A*02:01 is confirmed, converting MAGE-A4, PRAME and NY-ESO-1 from unassessable to live expression questions on one archival block. Leiomyosarcoma expression runs well below synovial sarcoma for MAGE-A4 and NY-ESO-1; PRAME is the likeliest of the three.

Three bundles carry thirteen of the fourteen actionable rows: a comprehensive tissue DNA plus RNA panel, which also settles TP53 versus clonal hematopoiesis; an IHC block led by INI1; the cancer/testis stain set, plus a blood draw for the germline HR panel. The fourteenth, FAP, defers to imaging. All ride the 2024 resection held at an outside institution, so block release is the rate-limiting step; with LMS-04 pending, serial ordering costs weeks she does not have. The INI1 stain would turn the plasma SMARCB1 frameshift from hypothesis to actionable, against a hard prior: 170 consecutive uterine smooth-muscle tumors all retained SMARCB1. Eight rows hand off to the target-validation track by survey ID. Most of these tests will come back negative; the case for each is the size of the option a positive opens, not the odds of one.

Gaps: biomarkers with no usable result on file

Microsatellite instability / mismatch-repair deficiency (MSI-H / dMMR)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

MSI-H and dMMR are read the same way whatever the primary site. Nothing in this record speaks to mismatch repair, and she becomes a checkpoint-blockade candidate the moment the planned first-line regimen stops working.

Mismatch-repair status is still unassessed: the plasma assay returned no MSI call, and no stain or PCR has been run on either specimen. Uterine leiomyosarcoma is nearly always mismatch-repair proficient, so a negative is the expected result, and the case for testing is that four stains on a block already being cut for INI1 close a tumor-agnostic PD-1 indication that would otherwise sit open at progression. Loss of any one protein would also carry Lynch implications for a woman in her thirties and for her first-degree relatives.

Recommended assay. MMR protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue (IHC · Archival FFPE; the September 2024 hysterectomy specimen has more tissue to spare, the 2026 lung core reflects current disease · 3-5 days)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)
- dostarlimab (PD-1 inhibitor · tumor-agnostic approval · NCT02715284)

Next step. Order now

References. nct:NCT02628067; nct:NCT02715284

Tumor mutational burden, high (TMB-H)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

TMB at or above 10 mutations per megabase supports pembrolizumab whatever the primary site. The Guardant plasma assay did not report a TMB value, so TMB here is unknown rather than low.

TMB is unreported: the plasma assay did not return one, and a plasma TMB at an unstated tumor fraction would not have been usable anyway. Leiomyosarcoma is copy-number driven and typically low-TMB, so a high value would be a surprise, but the tissue panel that reports it is the same order that settles MSI, the fusions, the HR genes, and whether the plasma TP53 call is tumor-derived. Ask explicitly for a validated comprehensive assay with a mutations-per-megabase figure, because a number off a small hotspot panel does not satisfy the indication.

Recommended assay. Validated comprehensive genomic profiling panel on tissue, reporting TMB in mutations per megabase (NGS_panel · Archival FFPE; the September 2024 uterine resection is preferred for tumor content and volume · 2-3 weeks)

Bundles with. Comprehensive tissue DNA plus RNA genomic profiling on archival tumor tissue, the single order that also returns TMB, MSI, NTRK and RET fusions, BRAF codon 600, the HR genes, the RB1 / ATRX / MED12 backbone of this histology, and tissue confirmation of the plasma TP53 and SMARCB1 calls

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)

Next step. Order now

References. nct:NCT02628067

B7-H3 (CD276) protein expression

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

B7-H3 sits on tumor cells and tumor vasculature across soft tissue sarcoma: 97% of specimens positive and about 69% high in a 153-patient series covering 15 subtypes, leiomyosarcoma among them. Of the surface targets on the panel, this is the one most likely to be positive here.

B7-H3 has still not been stained, and it carries the best odds of a positive result of any surface target in scope. Nothing is approved against it, so a positive buys trial access rather than a prescription, and the ADC, CAR-T and bispecific programs generally require documented expression at screening. Adding it to the block already being cut for INI1 and the MMR stains keeps a later trial referral from stalling while someone hunts for tissue.

Recommended assay. B7-H3 (CD276) IHC on tumor tissue (IHC · Archival FFPE; the 2026 lung core reflects current disease, the 2024 resection has more tissue and reads heterogeneity better · 1 week)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3

A positive result would open.

- ifinatamab deruxtecan (B7-H3-directed ADC · trial only)
- enoblituzumab (mAb · trial only)
- vobramitamab duocarmazine (ADC · trial only)

Next step. Bundle with planned testing

References. pmid:39478506

MAGE-A4 expression (HLA-A*02:01-restricted)

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

MAGE-A4 carries the first approved TCR-T therapy for a solid tumor, and this patient clears its restriction gate: allele-level typing confirms HLA-A*02:01. Expression has never been tested, and it is the only remaining half of the screening question.

The expensive half of this question is already answered: she is HLA-A*02:01 positive, so MAGE-A4 expression is the only thing standing between her and a class of agents she is a plausible candidate for. Expression in leiomyosarcoma runs well below the 70 to 80 percent seen in synovial sarcoma, so a negative is a realistic and useful outcome, and even a positive would not make her eligible for afamitresgene autoleucel, whose licence covers synovial sarcoma alone. The realistic route is a MAGE-A4 solid-tumor TCR-T trial, and the stain is what decides whether that door is open or shut.

Recommended assay. MAGE-A4 expression by IHC or RT-PCR on tumor tissue, scored per the candidate program's screening assay (IHC · Archival FFPE; RT-PCR tolerates lower tumor content than IHC scoring · 1-2 weeks)

Bundles with. Cancer/testis antigen IHC set (MAGE-A4, PRAME, NY-ESO-1) on one archival block, now unconditional because HLA-A*02:01 is confirmed

A positive result would open.

- afamitresgene autoleucel (MAGE-A4-directed TCR-T · approved, other tumor type)

- uzatresgene autoleucel (MAGE-A4-directed TCR-T · trial only)

Next step. Order if tissue available

References. pmid:38554725; pmid:36624315

PRAME expression (HLA-A*02:01-restricted)

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

PRAME is expressed in a subset of sarcomas including leiomyosarcoma, and the ImmTAC and TCR-T programs behind it enrol HLA-A*02:01-positive patients, which she is. Antigen expression has never been measured.

PRAME is now a straight expression question, because the HLA gate that used to make it unassessable is answered. Of the three cancer/testis antigens in scope it has the most credible chance of being positive in this histology, and both programs behind it enrol HLA-A*02:01-positive, PRAME-expressing solid tumors, so a positive buys trial access rather than a prescription. Reported expression in leiomyosarcoma runs below synovial sarcoma, so a negative stain would not be surprising and would close the axis cleanly.

Recommended assay. PRAME expression by validated IHC or RT-PCR at the threshold the candidate protocol specifies (IHC · Archival FFPE; RT-PCR tolerates lower tumor content than IHC scoring · 1-2 weeks)

Bundles with. Cancer/testis antigen IHC set (MAGE-A4, PRAME, NY-ESO-1) on one archival block, now unconditional because HLA-A*02:01 is confirmed

A positive result would open.

- brenetafusp (PRAME-directed bispecific TCR (ImmTAC) · trial only)
- IMA203 (PRAME-directed TCR-T · trial only)

Next step. Order if tissue available

References. pmid:40205198

BRAF V600E mutation

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

BRAF V600E is actionable across nearly all primary sites, colorectal excepted. It is a rarity in leiomyosarcoma, and it belongs on the list because any comprehensive tissue panel reports codon 600 whether or not anyone thought to ask.

No BRAF result is on file and the prior probability in uterine leiomyosarcoma is very low. No separate order is warranted: codon 600 is covered by the comprehensive tissue panel already recommended for TMB, MSI and the HR genes. Non-V600E class II and III BRAF alterations, should any turn up, do not carry this indication.

Recommended assay. Tumor NGS panel covering BRAF codon 600 (NGS_panel · Archival FFPE · 2-3 weeks)

Bundles with. Comprehensive tissue DNA plus RNA genomic profiling on archival tumor tissue, the single order that also returns TMB, MSI, NTRK and RET fusions, BRAF codon 600, the HR genes, the RB1 / ATRX / MED12 backbone of this histology, and tissue confirmation of the plasma TP53 and SMARCB1 calls

A positive result would open.

- dabrafenib plus trametinib (BRAF inhibitor plus MEK inhibitor · tumor-agnostic approval · NCT02034110)

Next step. Bundle with planned testing

References. nct:NCT02034110

HER2 protein overexpression (IHC 3+)

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

HER2 IHC 3+ supports trastuzumab deruxtecan irrespective of primary site. Strong HER2 staining is uncommon in leiomyosarcoma, but nothing on file addresses it and the stain is routine.

HER2 protein expression is untested. Only 3+ carries the tumor-agnostic indication and 3+ is unlikely in uterine leiomyosarcoma, so a 1+ or 2+ result should be reported as HER2-low and read as a weaker, less reliable signal rather than as a positive. A block is being cut for other stains anyway, which makes adding HER2 cheap and settles the question either way. The ERBB3 G914E variant on the plasma report sits at 0.9% VAF, near the assay limit of detection, and is not a reason to pursue the HER family.

Recommended assay. HER2 IHC on tumor tissue (IHC · Archival FFPE block · 1 week)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3; ERBB2 copy number also comes off the comprehensive tissue NGS panel

A positive result would open.

- trastuzumab deruxtecan (HER2-directed ADC · tumor-agnostic approval · NCT04482309)

Next step. Bundle with planned testing

References. nct:NCT04482309

RET gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

The RET fusion indication ignores primary site. RET fusions in uterine leiomyosarcoma are rare to the point of being anecdotal, but the RNA fusion assay ordered for NTRK answers RET in the same run.

RET fusion is untested and will almost certainly be negative in this histology, and the plasma report on file does not enumerate a RET result either way. It earns a row because it costs nothing extra once an RNA fusion panel is ordered for NTRK, and because a positive would put an oral targeted agent ahead of further cytotoxic chemotherapy. A RET point mutation is a different finding and does not carry this indication.

Recommended assay. RNA-based fusion NGS panel covering RET (NGS_panel · Archival FFPE · 2-3 weeks)

Bundles with. Comprehensive tissue DNA plus RNA genomic profiling on archival tumor tissue, the single order that also returns TMB, MSI, NTRK and RET fusions, BRAF codon 600, the HR genes, the RB1 / ATRX / MED12 backbone of this histology, and tissue confirmation of the plasma TP53 and SMARCB1 calls

A positive result would open.

- selpercatinib (selective RET inhibitor · tumor-agnostic approval · NCT03157128)

Next step. Bundle with planned testing

References. nct:NCT03157128

FAP (fibroblast activation protein) expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

FAP is the panel's stromal target, and mesenchymal tumors carry it on the malignant cells as well as on associated fibroblasts, which is why sarcoma shows up so reliably on FAPI PET. The panel's expression evidence comes from carcinoma stroma, so read sarcoma relevance as supported by imaging rather than by an IHC series in this histology.

FAP is untested, and no FAP-directed agent is approved anywhere, so a positive would matter only if a radioligand or bispecific trial were within reach. FAP IHC is not a routine clinical stain and most laboratories do not offer it, which makes speculative testing a poor use of a block that several higher-yield stains are already queued on. Revisit if a FAP-directed trial becomes a live option, at which point FAPI PET is usually the practical screening route and spends no tissue.

Recommended assay. FAP IHC on archival tissue, or FAPI PET where a FAP-directed program requires it (IHC · Archival FFPE · 1-2 weeks)

Bundles with. The archival IHC block if a FAP-directed program comes into play; otherwise the trial's own screening assay

A positive result would open.

- 177Lu-FAP-2286 (radioligand therapy · trial only)
- RO7300490 (bispecific (FAP-4-1BBL) · trial only)
- simlukafusp alfa (FAP-IL2v immunocytokine · trial only)

Next step. Defer

References. pmid:32601975

NY-ESO-1 (CTAG1B) expression, HLA-A*02:01-restricted

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

NY-ESO-1 is the third A*02:01-restricted cancer/testis antigen with a clinical-stage TCR-T program, and the restriction gate is satisfied. Its expression evidence sits in synovial sarcoma and myxoid liposarcoma rather than leiomyosarcoma, so the prior here is low.

NY-ESO-1 is an expression question now rather than an unassessable one, since HLA-A*02:01 is confirmed, but the prior probability in leiomyosarcoma is low. Letetresgene autoleucel is being developed in synovial sarcoma and myxoid/round cell liposarcoma, so even a positive stain would point at a solid-tumor cohort rather than a trial written for this histology. Add it to the cancer/testis block being cut for MAGE-A4 and PRAME rather than ordering it on its own.

Recommended assay. NY-ESO-1 expression by IHC or RT-PCR, run alongside MAGE-A4 and PRAME on the same block (IHC · Archival FFPE; RT-PCR is the more forgiving option at low tumor content · 1-2 weeks)

Bundles with. Cancer/testis antigen IHC set (MAGE-A4, PRAME, NY-ESO-1) on one archival block, now unconditional because HLA-A*02:01 is confirmed

A positive result would open.

- letetresgene autoleucel (NY-ESO-1-directed TCR-T · trial only · NCT03967223)

Next step. Bundle with planned testing

References. nct:NCT03967223

PD-L1 protein expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

PD-L1 gates no approval in soft tissue sarcoma, so this is supporting rather than decisive. It stays in scope because the pulmonary lesions were read as infiltrative and possibly inflammatory, which makes the immune phenotype a question someone will raise.

PD-L1 has not been stained and cannot be read off a plasma assay. In sarcoma it neither opens nor closes a labelled indication, so treat the result as supporting information for trial screening and for weighing any checkpoint-blockade proposal rather than as a decision point. If it is stained, use the lung metastasis and match the clone and cutoff to whichever agent is under consideration, because a result generated for one companion diagnostic does not transfer to another indication.

Recommended assay. PD-L1 IHC using the clone and scoring system matched to the intended agent (IHC · Archival FFPE; the 2026 lung core is the more informative specimen for current immune phenotype · 1 week)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3

A positive result would open.

- PD-1 / PD-L1 inhibitors (checkpoint inhibitor · trial only)

Next step. Bundle with planned testing

AXL receptor tyrosine kinase expression

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not measured

AXL marks mesenchymal and EMT-high tumor cells, which is the constitutive state of a sarcoma rather than an acquired one, and it tracks with chemotherapy resistance. The panel's expression evidence comes from carcinomas, so read this as plausible in leiomyosarcoma rather than established.

AXL has not been assessed, and no AXL-directed agent is approved: enapotamab vedotin and tilvestamab both stopped at completed early-phase studies. Testing would not change anything she can access now or at progression. Revisit only if an AXL-directed program opens with an expression-based entry criterion.

Recommended assay. AXL IHC on tumor tissue (IHC · Archival FFPE · 1-2 weeks)

Bundles with. Would ride on the archival IHC block if an AXL-directed program required documented expression

A positive result would open.

- enapotamab vedotin (ADC · trial only)
- tilvestamab (mAb · trial only)

Next step. Defer

Measured, but not to decision resolution

A result exists for each of these, but not at the resolution a treatment decision needs. The workup that would close each gap is carried into the Target validation paths report.

NTRK1 / NTRK2 / NTRK3 gene fusion

Priority: essential

On file. profile.json:biomarkers[13] NTRK fusion: 'not settled by the Guardant plasma panel (fusion detection in plasma is insensitive at low tumor fraction)', confirmation_status 'ngs_pending'; no RNA-based fusion panel and no tissue DNA NGS with fusion calling on file; profile.json:organ_function.note_ctdna_tumor_fraction: plasma tumor fraction was not supplied, so non-detection on the plasma assay is non-exclusionary

Gap. All that is on file is non-detection on a plasma DNA assay run at an unstated tumor fraction. Plasma fusion calling is insensitive when shedding is low, and DNA capture misses breakpoints sitting in large NTRK introns. An RNA-based fusion panel on archival tissue, covering NTRK1, NTRK2 and NTRK3, is what would let a negative be read as a negative.

A plasma assay reported no NTRK fusion, which is not the same thing as a negative: tumor fraction was never supplied, and plasma DNA fusion calling misses breakpoints an RNA assay would catch. Prevalence in uterine leiomyosarcoma is very low, so a true negative is the expected outcome, and what justifies the assay is that a positive turns a chemotherapy-only landscape into an oral TRK inhibitor with high and durable response rates. It rides on the same tissue order that answers RET, BRAF, TMB and MSI, so it adds little beyond the tissue itself.

Homologous recombination deficiency / BRCA1 / BRCA2 pathogenic variant

Priority: essential

On file. profile.json:biomarkers[12] Homologous recombination status incl. BRCA2: 'not settled by the Guardant plasma panel (no BRCA1/2 alteration reported, but plasma sensitivity is limited at low tumor fraction and HRD signatures require tissue)', confirmation_status 'ngs_pending'; no tissue HRD score and no germline multigene panel on file; profile.json:organ_function.note_ctdna_tumor_fraction: plasma tumor fraction was not supplied, so non-detection on the plasma assay is non-exclusionary

Gap. The plasma panel reported no BRCA1/2 alteration, at an unstated tumor fraction and with no genomic-instability score, and no germline testing has been done. Closing this needs tissue NGS across the HR genes with an HRD score, plus a germline multigene panel on blood.

The plasma assay does not settle homologous-recombination status: an absent BRCA1/2 call at unknown tumor fraction is weak evidence, and an HRD signature cannot be derived from that assay at all. Roughly a tenth of uterine leiomyosarcomas carry an HR pathway mutation with a larger fraction functionally deficient by RAD51 assay, and a pathogenic finding would put olaparib plus temozolomide and PARP-inhibitor trials on the table for later lines. Order the germline panel alongside the tumor test rather than after it, since somatic-only testing leaves both her own risk and her relatives' screening unanswered.

Already established

- **HLA class I genotype** — profile.json:biomarkers[4] HLA class I typing (allele-level): A*02:01, A*03:01; B*39:01, B*44:02; C*05:01, C*12:03, confirmation_status 'confirmed'; profile.json:targetable_features[2] carries the A*02:01 result as an opened axis pending antigen expression

Panel targets out of scope for this tumor

66 of 72 panel targets were screened and set aside as having no plausible connection to this patient's tumor type or stated features: ALPPL2 · ANTXR1 · CA9 · CD19 · CD200 · CD37 · CD46 · CD70 · CDCP1 · CDH17 · CDH3 · CDH6 · CEACAM5 · CLDN18 · CLEC12A · CT83 · DLK1 · DLL3 · EGFR · EGFR (variant III) · ENPP3 · EPCAM · EPHA2 · EPHB4 · ERBB2 · ERBB3 · F3 · FOLH1 · FOLR1 · FZD7 · GPC3 · GPRC5D · GUCY2C · HAVCR1 · IL13RA2 · IL3RA · ITGA3 · ITGB6 · L1CAM · LGR5 · LILRB4 · LY6E · LYPD3 · MAGEA1 · MAGEC2 · MERTK · MET · MMP14 · MSLN · MUC1 · NECTIN4 · PODXL · PSCA · PTK7 · ROR1 · SEZ6 · SLC34A2 · SLC44A4 · SLITRK6 · SSTR2 · SSX2 · STEAP1 · TACSTD2 · TNFRSF10B · TNFRSF17 · TSPAN8